HIGH ACRYLONITRILE NBR PATENT RESEARCH REPORT

1. ABSTRACT

This report provides a concise technical summary of the research scope and target compound High Acrylonitrile NBR, analyzing 1 primary patent and no competitor patents or website sources. The major technical approaches observed include the use of metathesis reactions and various parameters such as temperature, pressure, and catalysts. Key process trends identified include the importance of temperature control and the use of specific catalysts.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: CN103403034 - Patent Title: Functionalized nitrile rubber and its preparation - Assignee: Not disclosed - Jurisdiction: CN - Publication Year: 2016-05-04 - Material Type: Nitrile rubber - Relevance to target: Relevant to High Acrylonitrile NBR

Polymerization Method - Nitrile: 1 g (Used in the metathesis reaction) - Temperature: 20 °C (Reaction temperature) - Under: 5 hours (Reaction time) - Up: 50 % (Partial hydrogenation) - Cocatalysts: 33 parts by weight (Based on 100 parts by weight of the nitrile rubber) - But: 2 parts by weight (Based on 100 parts by weight of the nitrile rubber) - Cocatalyst: 5 parts by weight (Based on 100 parts by weight of the nitrile rubber) - Temperature: 100 °C (Reaction temperature) - Pressure: 150 bar (Reaction pressure)

Relevant Experimental Evidence - The reaction was started by adding 50 mg (2.95*10^-4 mol) of the metathesis catalyst to a solution of 1 g of nitrile rubber in 20 ml of dichloromethane. - The reaction mixture was then stirred at room temperature for 5 hours. - The yield of the reaction was 2.4 g (17.8 mmol, 58%).

Technical Relevance This patent is relevant to High Acrylonitrile NBR as it describes a method for the preparation of functionalized nitrile rubber, which is a key component of High Acrylonitrile NBR.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Trend Analysis: - The use of metathesis reactions is a common technical approach in the preparation of High Acrylonitrile NBR. - Temperature control is an important parameter in the preparation of High Acrylonitrile NBR.

4. CONCLUSION

The preparation of High Acrylonitrile NBR involves the use of metathesis reactions and various parameters such as temperature, pressure, and catalysts. The use of specific catalysts and temperature control are key factors in the preparation of High Acrylonitrile NBR.

5. REFERENCES

- CN103403034 - Functionalized nitrile rubber and its preparation